Code

MACT 4232 (Time Series) - Assignment 1

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This is an R Markdown (http://rmarkdown.rstudio.com) Notebook. When you execute code within the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor inside it and pressing *Cmd+Shift+Enter*.

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```
setwd("/Users/omar/desktop")
getwd()
```

```
[1] "/Users/omar/Desktop"
```

Question 1: Use R to plot the time series and comment on any trends or other features of the data

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```
df = scan("global_temps.dat")
```

```
Read 20 items
```

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```
head(df)
```

```
[1] -0.09 -0.22  0.11  0.04  0.10  0.19
```

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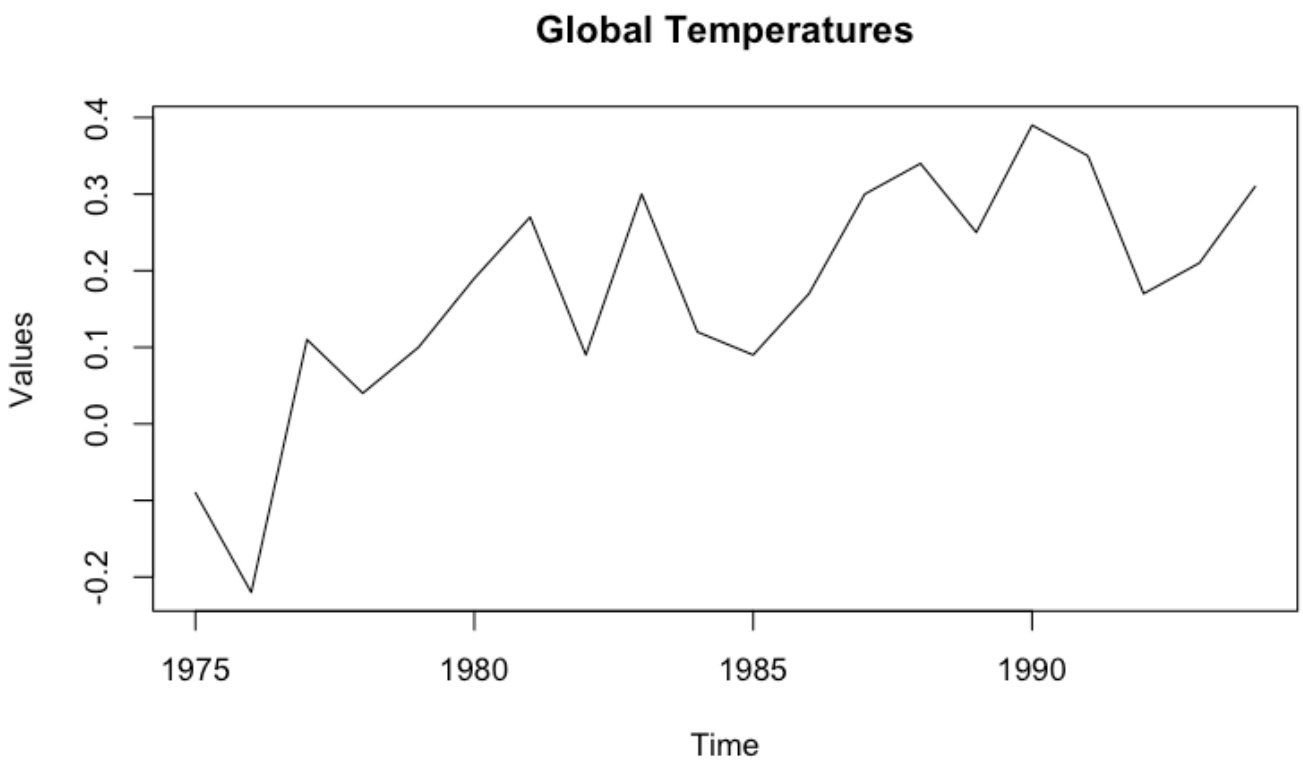
```
# 'ts' is short for time series

# frequency = 1 because we're dealing with ANUAL data

# start = c(1975) because the starting year is 1975 and the starting month is unknown

df_ts = ts(df, frequency = 1, start = c(1975))

# Plot the time series data
plot.ts(df_ts, main = "Global Temperatures", xlab = "Time", ylab = "Values")
```



Comments on the trends & other features of the data:

1. The fluctuations shown in the time series plot are not relatively constant, indicating that this particular data is more suitable for the multiplicative model rather than the additive model.
2. The data also exhibits an overall increasing trend, indicating that global temperatures increase over time. With that said, this particular trend is not perfectly linear due to the noticeable fluctuations.
3. There are a few clear peaks and dips, which could be reflections of specific climate events or anomalies which lead to these temporary deviations from the overall trend.

Question 2: Use R to plot the series and a 4-point centered moving average. Is there any trend?

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```
df1 = scan("quarterly_arrivals_new_zealand.dat")
```

```
Read 16 items
```

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```
head(df1)
```

```
[1] 239 204 238 253 238 193
```

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```
# frequncy = 4 because we're dealing with QUARTERLY data

# start = c(1990, 1) because the starting year is 1990 and the starting month is Marc
h which falls in the FIRST quarter of the year

df1_ts = ts(df1, frequency = 4, start = c(1990, 1))
```

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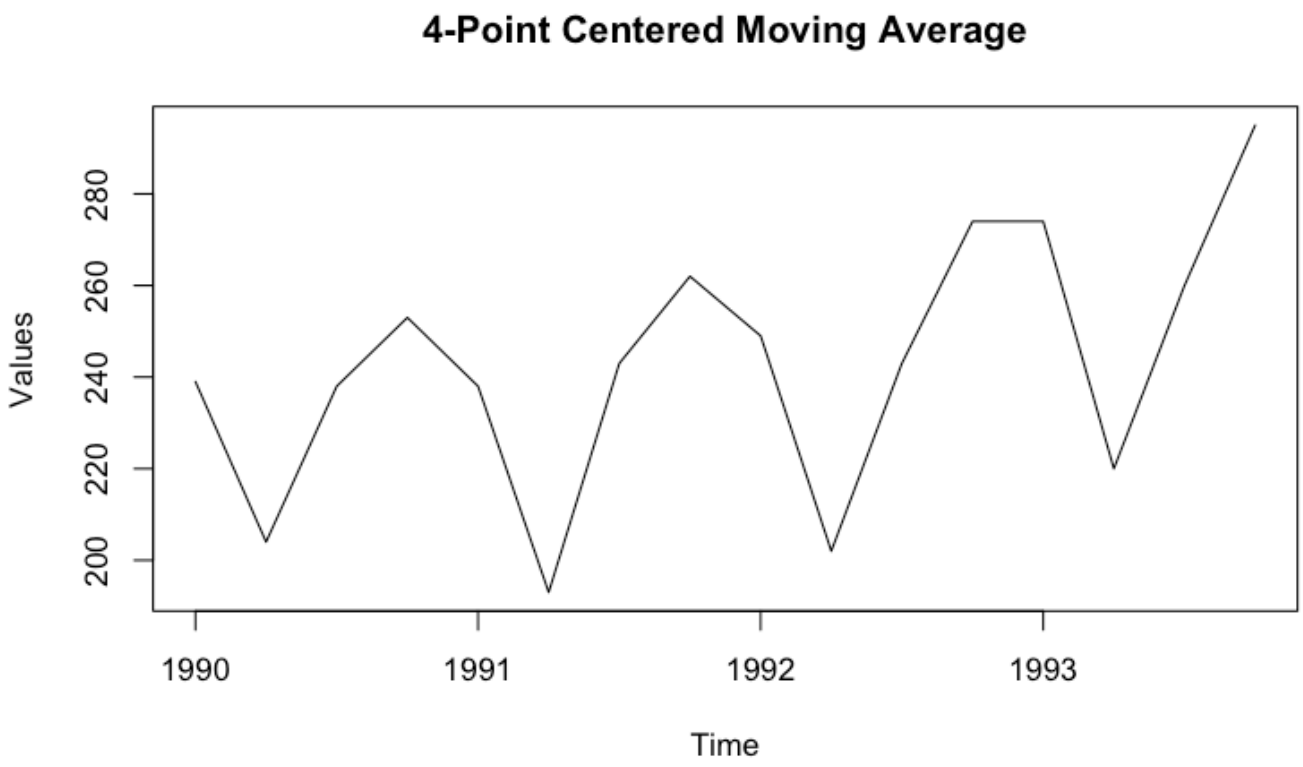
```
# To get the Centered Moving Average (CMA) for quarterly data we use the following:

# The standard for CMA is that sides = 2

# The first argument of 'filter' is equal to 1/(2*n), so in this case with quarterly
data, it's 1/(2*4) which is 1/8

n = 4 # this is also because we're dealing with quarterly data

df1_ts.ma = filter(df1_ts, filter = c(1/2*n, rep(1/n, n-1), 1/2*n), sides = 2)

plot.ts(df1_ts, main = "4-Point Centered Moving Average", xlab = "Time", ylab = "Valu
es")
```



Is there any trend?

Yes, there is indeed a trend. There is a cyclical oscillation with an upward shift in the peaks and troughs. The repeated sets of peaks and troughs indicate a potential seasonal or cyclical component where the peaks seem to be taking place in the second half of the year and the troughs take place in the first quarter of the year. Additionally, the increasing amplitude of oscillations over time may indicate growing variability in the data.

Add a new chunk by clicking the *Insert Chunk* button on the toolbar or by pressing *Cmd+Option+I*.

Question #3:

(a) Use the moving average to estimate the trend

Quarter	(y_t)	4-point	2-point
1996 (1)	18	—	
(2)	30	—	
(3)	44	31	
(4)	32	37.75	34.375
1997 (1)	45	45.75	41.75
(2)	62	45.25	45.50
(3)	42	52.25	48.75
(4)	60	61.50	56.875
1998 (1)	82	59	60.25
(2)	52	67.25	63.125
(3)	75	77.25	72.25
(4)	100	—	

(b) Compute the seasonal factors

$$\text{Seasonal Irregular Component} = \frac{\text{Value}}{\text{CMA}}$$

Year	Quarter	Value	CMA	Seasonal Irregular Component
96	3	44	34.375	1.28
96	4	32	41.75	0.7667
97	1	45	45.50	0.9890
97	2	62	48.75	1.2718
97	3	42	56.875	0.7385
97	4	60	60.25	0.9959
98	1	82	63.125	1.2990
98	2	52	72.25	0.7197

$\Rightarrow$ $\boxed{\text{Quarter 1:}}$ $\dfrac{0.989 + 1.299}{2} = 1.144$

$\boxed{\text{Quarter 2:}}$ $\dfrac{1.2718 + 0.7197}{2} = 0.996$

$\boxed{\text{Quarter 3:}}$ $\dfrac{1.28 + 0.7385}{2} = 1.0093$

$\boxed{\text{Quarter 4:}}$ $\dfrac{0.7667 + 0.9959}{2} = 0.8813$

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Question #4.

What's the exp. smoothed value for the next period if the forecast & actual values are 12 & 15 and $\alpha = 0.7$?

Single exp. smoothing : $F_{t+1} = \alpha X_t + (1-\alpha) F_t \Rightarrow X_t = 15,\ F_t = 12,\ \alpha = 0.7$

$$\Rightarrow F_{t+1} = 0.7(15) + (1-0.7)(12) = 10.5 + 3.6 \boxed{= 14.1}$$

Question #5.

What'll be the forecast value of the # of rifles to be ordered in the next month if defense management ordered 120 rifles & the forecasted value for this month is 115, and $\alpha = 0.8$?

Forecast value for period $t+1 = F_{t+1} = ?$

$\Rightarrow$ Single exp. smoothing : $F_{t+1} = \alpha X_t + (1-\alpha) F_t \Rightarrow X_t = 120,\ F_t = 115,\ \alpha = 0.8$

actual value for this month

forecast value for this month

$$\Rightarrow F_{t+1} = 0.8(120) + (1-0.8)(115)$$

$$\Rightarrow F_{t+1} = 96 + 23 \boxed{= 119}$$

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